- Homework
  - Problem 1
  - Problem 2
  - Problem 3

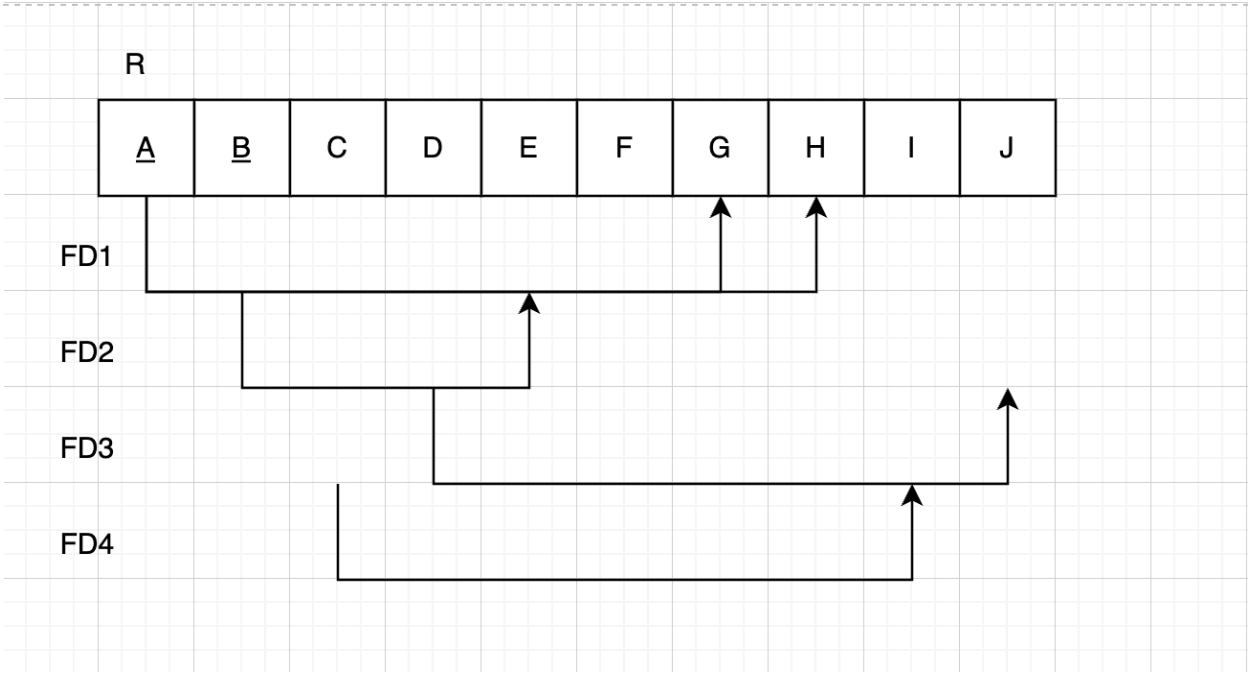
Homework

A homework for class **MET CS 579** at Boston University, instructed by **Jae Lee**. Student Jin Yu Zhang submitted on **April 12, 2023**.

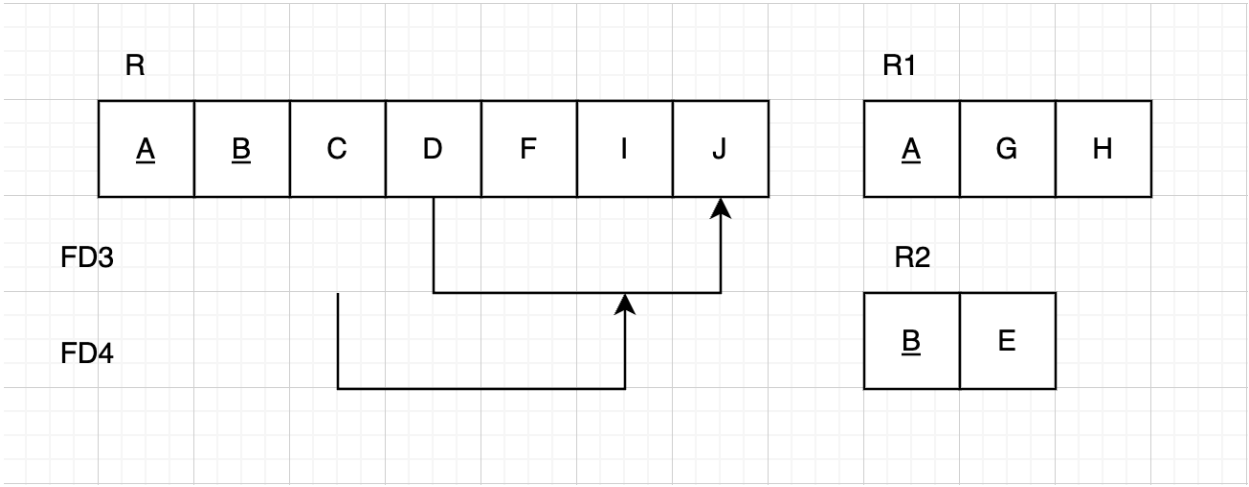
Problem 1

- (a): Doesn't exist. There are two duplicated value `10`, and they map to different values `b1` and `b2`
- (b): FD may hold
- (c): Doesn't exist. There are two duplicated value `b1`, and they map to different values `10` and 13
- (d): Doesn't exist. There are three duplicated value `c1`, and they map to different values `10`, 11, and 13
- (e): FD may hold

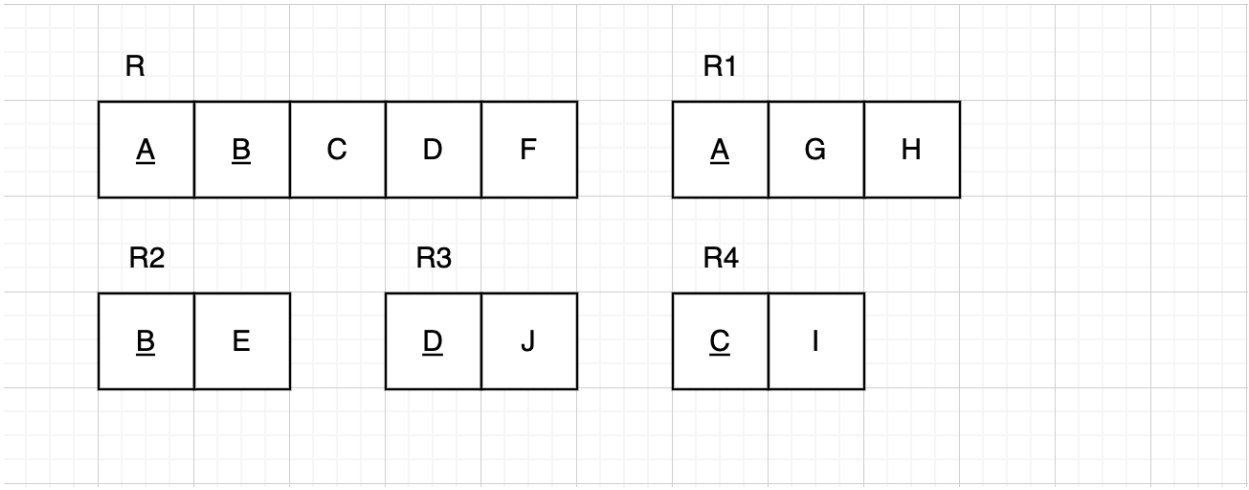
Problem 2



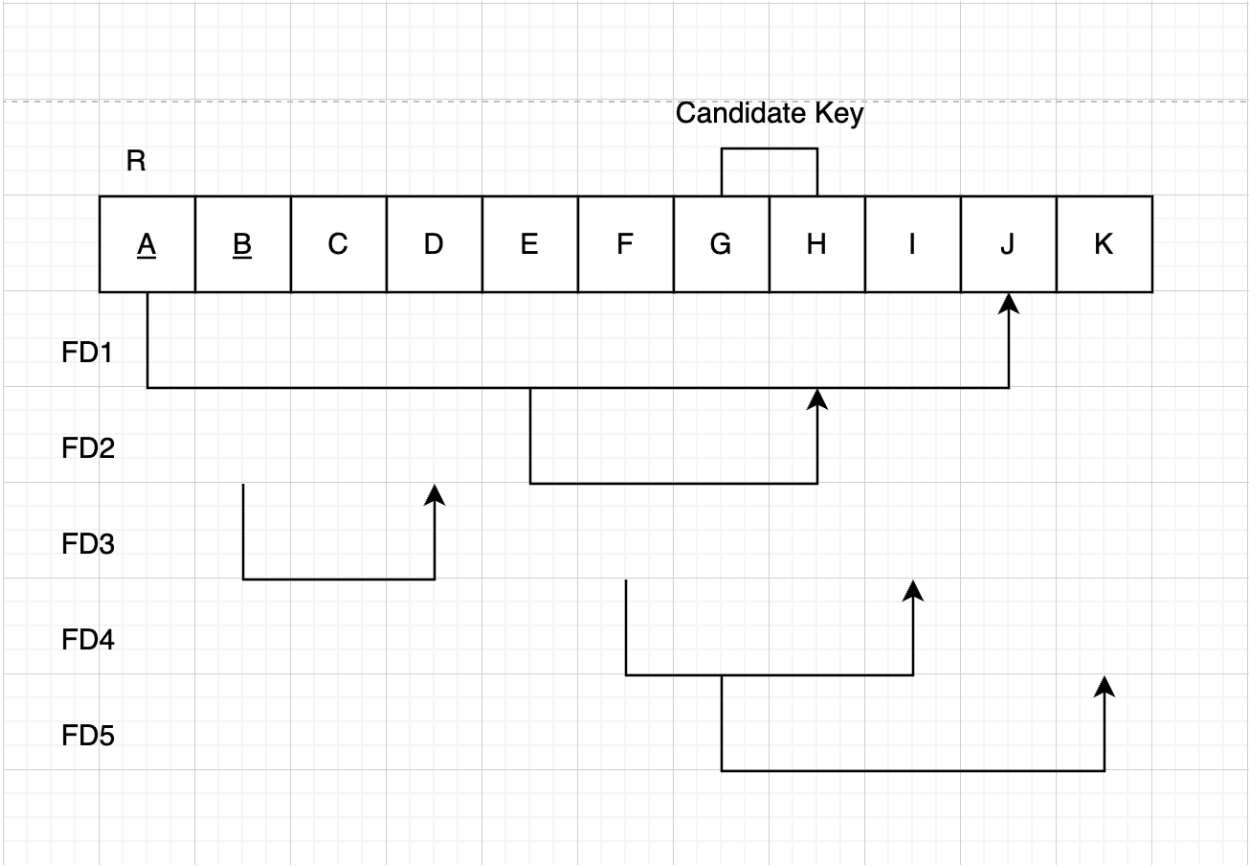
Normalize R to 2NF. Primary key attributes as underlined. A from R1 is a foreign key of A from R, B from R2 is a foreign key of B from R:



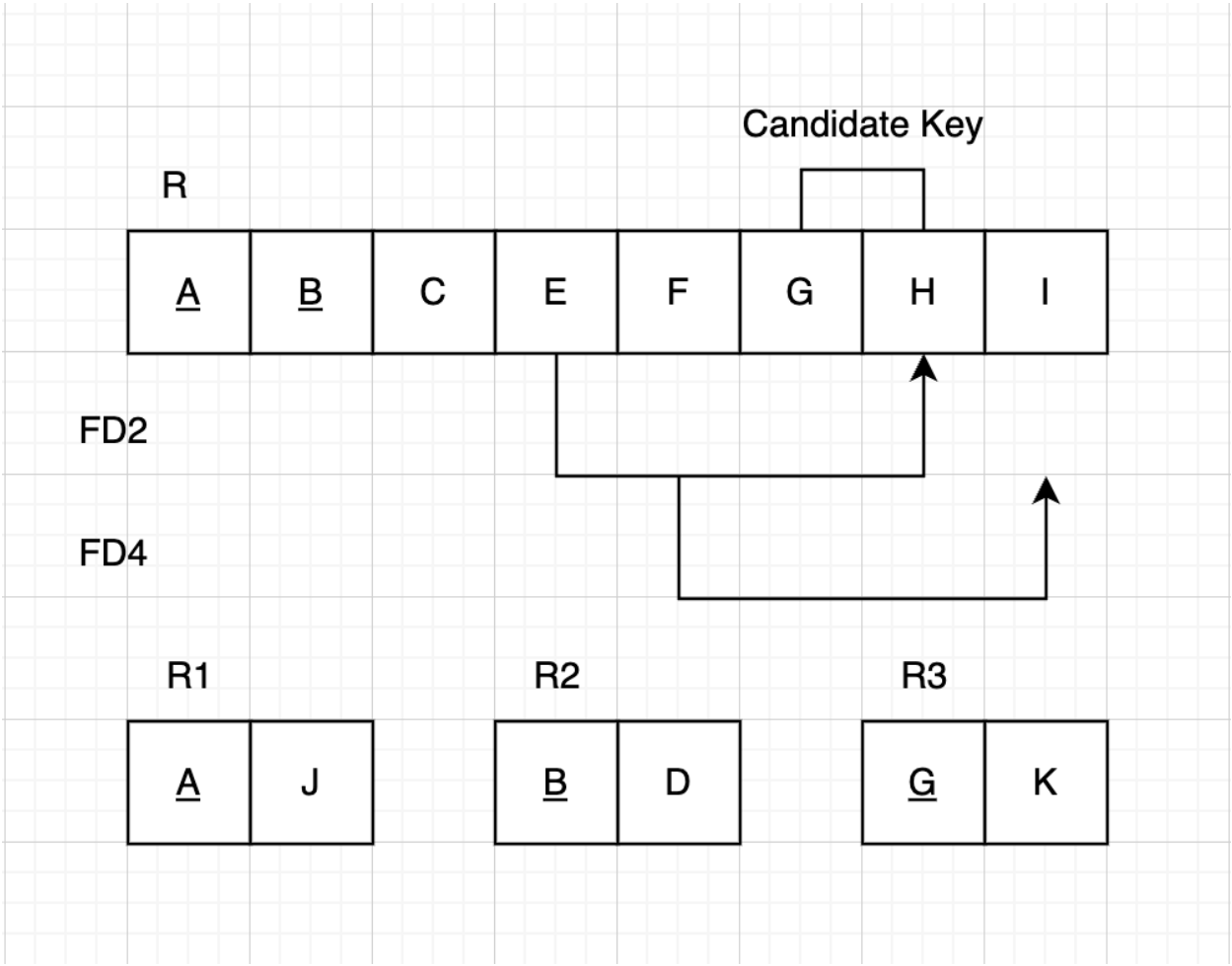
Normalize R to 3NF. Primary key attributes as underlined. A from R1 is a foreign key of A from R, B from R2 is a foreign key of B from R, D from R3 is a foreign key of D from R, C from R4 is a foreign key of C from R:



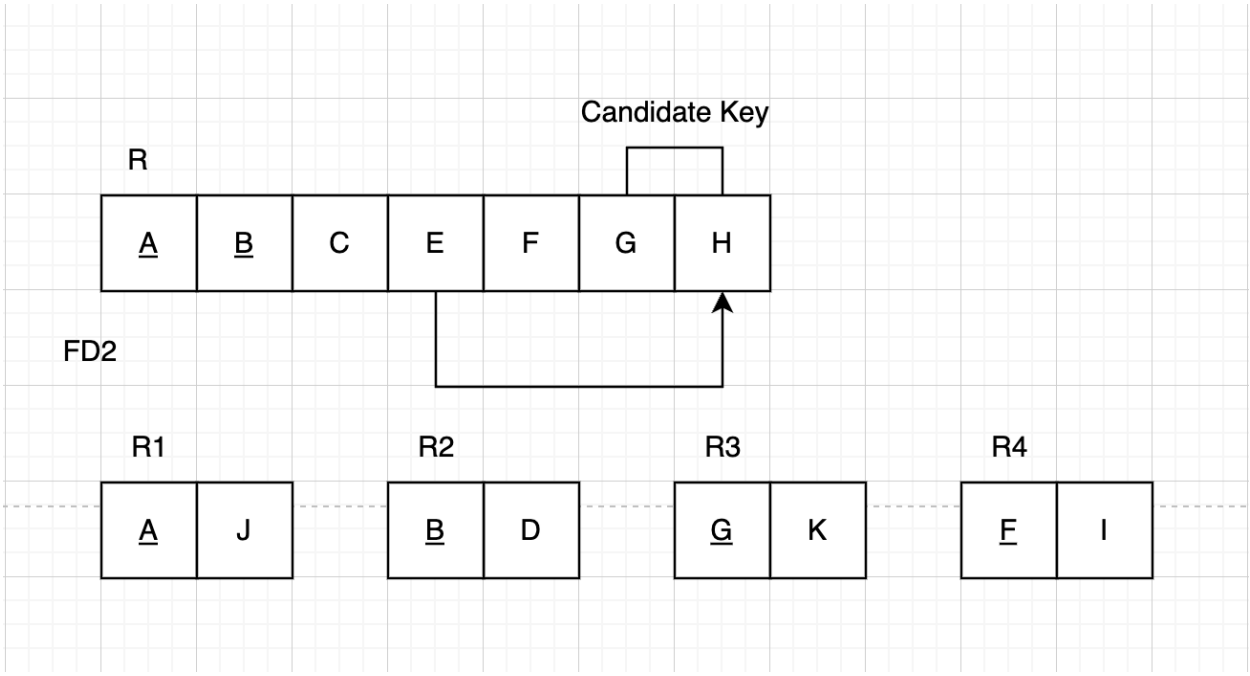
Problem 3



Normalize R to 2NF. Primary key attributes as underlined. A from R1 is a foreign key of A from R, B from R2 is a foreign key of B from R, G from R3 is a foreign key of G from R:



Normalize R to 3NF. Primary key attributes as underlined. A from R1 is a foreign key of A from R, B from R2 is a foreign key of B from R, G from R3 is a foreign key of G from R, F from R4 is a foreign key of F from R:



Normalize R to BCNF. Primary key attributes as underlined. A from R1 is a foreign key of A from R, B from R2 is a foreign key of B from R, G from R3 is a foreign key of G from R, F from R4 is a

foreign key of F from R, E from R5 is a foreign key of E from R:

